

NeuronFabric: A Software Reference Architecture for On-Chip Transformer Training with Local Adam

BF16W Weights, Vocabulary Budget, and a Path to FPGA Training without a Host CPU

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Abstract

Every neural chip we are aware of either does inference only (Groq LPU, Apple ANE, IBM NorthPole) or ships gradients off-chip to a host GPU or CPU for the weight update (Cerebras, Tenstorrent). The weight update never happens on the chip that computed the activations.

This paper describes an attempt to fix that. We built a complete C# software prototype of a transformer that runs forward pass, backpropagation, and Adam weight update entirely in one process with no external framework. The purpose is to establish that the math works and the numbers are right, before committing to silicon.

The model we trained is a 334K parameter autoregressive transformer ($d=88$, $H=4$ heads, $f=264$, $L=4$ layers, vocab= 256) trained on the Shakespeare corpus. The BF16W variant reaches eval loss 1.5477 at 80K samples (GPU FP32 oracle: 1.5226), with coherent character-level text generation confirmed.

We also introduce **BF16W**: storing weights in BF16 while keeping Adam moments in FP32. This is not a training approximation — it is a hardware necessity. A 334K FP32 model with Adam moments requires 4.0 MB — exactly the ZCU102 BRAM limit, leaving zero headroom for activation buffers. With BF16W it requires 3.34 MB, leaving 660 KB free.

We describe the vocabulary-budget constraint we discovered during earlier experiments, quantify the BF16W SRAM saving, and lay out the FPGA training target as the next step. No FPGA measurements are included in this paper; FPGA implementation and measurement are left for future work.

Code: github.com/Binoculars-X/neuro-fabric (release v1.0.3, commit 97c09f3).

Publication purpose: This paper serves as a public architectural disclosure and software reference implementation for future FPGA/ASIC implementations of the NeuronFabric system, establishing prior art for the local Adam update architecture.

1 Introduction

A human brain learns continuously at roughly 20 W. Training large language models at the scale of GPT-4 has been estimated at tens of megawatts sustained over months [1] — a factor of over a million in energy per unit of capability gained relative to biological learning. That gap is not mainly about transistor count or clock speed. It is about *where the weight update happens*.

The brain updates synaptic weights locally, in place, with no external optimizer and no weight traffic across a bus. Every neural accelerator we are aware of does the opposite: it separates the

compute from the update, ships gradients off-chip, and runs an optimizer on a host CPU or GPU. The chip that did the work does not do the learning.

The gap this creates is concrete:

- **Weight traffic.** GPU inference loads all weights from HBM on every forward pass. For a 70B model that is ~ 140 GB of memory traffic per sample.
- **Training power.** No existing ASIC supports on-chip weight updates. Training always requires external hardware.
- **No continuous learning.** An inference chip cannot update from new data without sending everything to an external GPU cluster and reloading — a cycle that takes minutes to hours even for fine-tuning.

NeuronFabric is an attempt to close that gap. The long-term goal is a silicon chip where every neuron holds its own weights and runs Adam locally — weights update in place, no gradient traffic off-chip, scaling achieved by connecting chips via activation links only. This paper establishes the software proof that this is numerically viable. FPGA implementation of the same training loop is the immediate next step.

1.1 Contributions

1. A complete C# implementation of a Pre-LN transformer with full backpropagation and local Adam, with no external ML framework dependency. Training, evaluation, and checkpointing run in a single binary.
2. **BF16W** — a mixed-precision scheme where weights are stored as BF16 (2 bytes) while Adam moments m and v remain FP32 (4 bytes each). We show this reduces SRAM from 12 bytes/parameter to 10 bytes/parameter with a negligible convergence penalty (+0.025 val loss) on Shakespeare.
3. The **vocabulary-budget constraint**: at small parameter budgets, the embedding table dominates the model and the transformer layers have almost nothing left to learn with. We quantify this across three domains and give the design implication for fixed-budget hardware.
4. A concrete FPGA training target for future work: the 334K BF16W Shakespeare model (vocab= 256) fits entirely in ZCU102 on-chip SRAM and can train without touching DDR.

1.2 Related Work

Inference-only ASICs. Groq LPU [3], Apple ANE [4], and IBM NorthPole [5] achieve excellent inference efficiency through SRAM-local weight storage and dataflow architectures. None support on-chip gradient computation.

Training accelerators. Cerebras WSE-3 [6] and Tenstorrent Wormhole [7] support training but the weight update still happens off-chip — gradients are aggregated on a host CPU or external optimizer.

Neuromorphic chips. Intel Loihi 2 [8] and SpiNNaker [9] support local weight updates via spike-timing-dependent plasticity (STDP) — a Hebbian rule that does not implement gradient descent. STDP cannot reproduce transformer training results.

2 Architecture

The software architecture is structured so that every abstraction has a direct hardware analogue. The hierarchy is:

$$\text{NeuronCore} \subset \text{AttentionCore} \subset \text{AttentionLayer} \subset \text{TransformerBus}$$

2.1 NeuronCore: Weight Storage + Compute Unit

A **NeuronCore** holds its own weight vector $\mathbf{w} \in \mathbb{R}^n$ and bias b . On the forward pass it computes a dot product; on the backward pass it writes an updated weight in place. There is no weight bus between cores — each core owns its weights permanently.

Forward:

$$z = \mathbf{w}^\top \mathbf{x} + b, \quad y = \sigma(z) \quad (1)$$

Backward and weight update (Adam):

$$\delta = g \cdot \sigma'(z) \quad (2)$$

$$m \leftarrow \beta_1 m + (1 - \beta_1) \delta \mathbf{x} \quad (3)$$

$$v \leftarrow \beta_2 v + (1 - \beta_2) (\delta \mathbf{x})^2 \quad (4)$$

$$\hat{m} = m / (1 - \beta_1^t), \quad \hat{v} = v / (1 - \beta_2^t) \quad (5)$$

$$\mathbf{w} \leftarrow \mathbf{w} - \eta \cdot \hat{m} / (\sqrt{\hat{v}} + \epsilon) \quad (6)$$

The weight update is a direct in-place write. On FPGA this maps to a BRAM write on the **Backward** control signal — no off-chip traffic.

BF16W variant: $\mathbf{w}$ is stored as **ushort** (BF16), cast to FP32 for computation, and cast back after the Adam step. Moments m and v remain FP32. This is the configuration used for all experiments in this paper.

2.2 Transformer Architecture (334K Shakespeare Config)

Hyperparameter	Value
Embedding dim d	88
Attention heads H	4
Head dim $d_h = d/H$	22
Feedforward dim f	264 ($= 3d$)
Transformer layers L	4
Sequence length T	128
Vocabulary	256 (byte-level)
Total parameters	334K

Table 1: 334K Shakespeare model — canonical configuration

Each transformer layer uses Pre-LN residual connections [2]:

$$x' = x + \text{MultiHeadAttn}(\text{LN}_1(x)) \quad (7)$$

$$x'' = x' + \text{FF}(\text{LN}_2(x')) \quad (8)$$

The feedforward block uses GeLU activation: $\text{FF}(x) = W_2 \cdot \text{GeLU}(W_1 x)$, where $W_1 \in \mathbb{R}^{d \times f}$ and $W_2 \in \mathbb{R}^{f \times d}$.

Weight tying: The output projection reuses the embedding matrix transposed: $\text{logits}[t, v] = \text{layerOut}[t] \cdot E[v]$. This removes $|V| \times d = 256 \times 88 = 22,528$ parameters from the output head with no accuracy cost — an important saving at small budgets.

2.3 Parameter Count Breakdown

Component	Formula	Params
Embedding	$ V \times d = 256 \times 88$	22,528
Attention per layer	$4 \times d \times d_h \times H = 4 \times 88 \times 22 \times 4$	30,976
Feedforward per layer	$2 \times d \times f = 2 \times 88 \times 264$	46,464
LayerNorm (approx)	~ 200 per layer	800
Per layer total		77,440
$\times 4$ layers		309,760
Embedding (weight-tied)		22,528
Total		$\sim 334\text{K}$

Table 2: Parameter breakdown for 334K Shakespeare model

2.4 Hardware Mapping

The design intent is that every software abstraction maps directly to a hardware block:

Software	Hardware target
<code>NeuronCore</code> weight array	BRAM block
<code>NeuronCore.Forward()</code>	DSP48 multiply-accumulate chain
<code>NeuronCore.Backward()</code>	BRAM write (same block, no bus)
<code>NeuronLayer</code> parallel dispatch	Parallel <code>always</code> blocks in Verilog
Softmax <code>exp()</code>	256-entry LUT in BRAM [†]
<code>TransformerBus</code> control enum	3-bit on-chip control signal [‡]

Table 3: Software-to-hardware mapping (FPGA target). [†]256 entries cover the stable-softmax argument range (attention logits after max-subtraction are bounded, typically $[-30, 0]$); range decomposition or interpolation may be required for full BF16 accuracy — deferred to FPGA implementation. [‡]Control signal is internal to the chip; host interface design (SPI, AXI-Lite, or similar) is deferred to FPGA implementation.

The control bus uses six states: `Idle` / `Encode` / `Forward` / `Backward` / `WeightRead` / `WeightWrite` — a 3-bit signal that drives all compute blocks. This is not an abstraction convenience; it is the intended silicon interface.

3 BF16W: Why Weight Compression Is Not Optional for FPGA

The FPGA training target requires all weights *and* Adam moments to live in on-chip SRAM during training. The Xilinx Zynq UltraScale+ ZCU102 has 32.1 Mb (≈ 4.0 MB) of BRAM. The 334K model in FP32 with Adam moments needs:

$$334,000 \times (4 + 4 + 4) \text{ bytes} = 334,000 \times 12 = 4.00 \text{ MB}$$

That barely fits. Adding activation buffers for backpropagation (~ 180 KB per layer, recomputed layer-by-layer) pushes the FP32 total over the limit.

BF16W drops weight storage from 4 bytes to 2 bytes:

$$334,000 \times (2 + 4 + 4) \text{ bytes} = 334,000 \times 10 = 3.34 \text{ MB}$$

This gives 660 KB of headroom for activation buffers, control state, and the embedding SRAM. The model fits comfortably.

Precision	Weights	Moments ($m+v$)	Total
FP32 Adam	1.34 MB	2.67 MB	4.00 MB
BF16W Adam	0.67 MB	2.67 MB	3.34 MB
ZCU102 BRAM			4.00 MB

Table 4: SRAM budget for 334K model. FP32 Adam fills the ZCU102 BRAM exactly with no headroom for activations. BF16W Adam leaves 660KB free — sufficient for activation buffers and control state.

BF16W implementation. Weights are stored as `ushort` arrays. Each training step: cast BF16 $\rightarrow$ FP32, compute forward + backward in FP32, apply Adam update in FP32, round FP32 $\rightarrow$ BF16, write back. Moments m and v stay in FP32 throughout — this is where precision matters most for Adam stability. The weight values themselves tolerate BF16 range (~ 3 decimal digits) because the Adam step size is small relative to weight magnitude.

Convergence. On the 80K-sample Shakespeare runs, CPU BF16W reaches eval loss 1.5477 versus GPU FP32 1.5226 — a gap of +0.025. Both variants converge qualitatively identically; the residual gap is consistent with the difference in compute precision and random seed (Section 5).

4 The Vocabulary-Budget Constraint

During earlier experiments with smaller models (100K parameters), our results were consistently worse than expected — not because backpropagation was wrong (the gradient checks passed), but because the embedding table was consuming most of the parameter budget before the transformer layers had any say.

The constraint. For a model with total parameters P , embedding dimension d , and vocabulary size $|V|$, the remaining capacity for transformer reasoning is:

$$P_{\text{reason}} = P - |V| \cdot d \tag{9}$$

We call the term $|V| \cdot d$ the **vocabulary tax**. With weight tying, the output projection costs nothing extra — but the embedding itself is unavoidable.

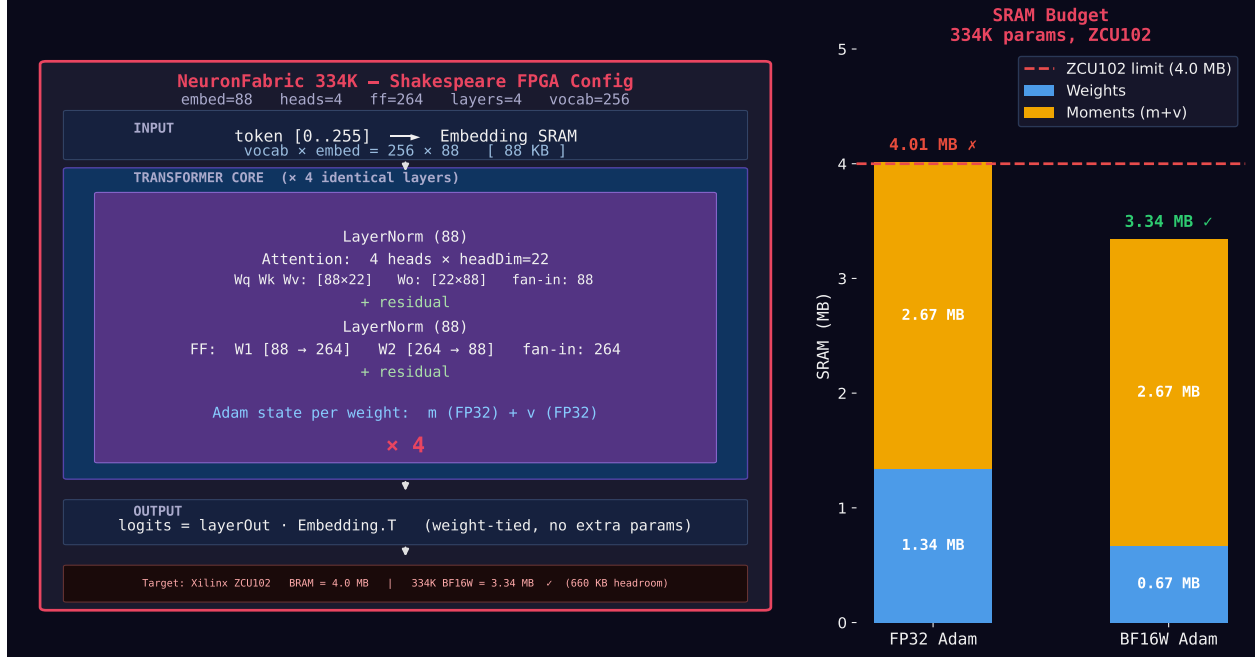


Figure 1: NeuronFabric 334K Shakespeare FPGA configuration (embed=88, heads=4, ff=264, layers=4, vocab=256). **Left:** chip architecture — token embedding SRAM (88 KB), four Pre-LN transformer layers each with local Adam state, weight-tied output projection. **Right:** SRAM budget on ZCU102 (4.0 MB BRAM). FP32 Adam requires 4.00 MB (no headroom for activations); BF16W Adam requires 3.34 MB, leaving 660 KB free. Weights: FP32 = 1.34 MB, BF16W = 0.67 MB. Moments ($m+v$, FP32 in both) = 2.67 MB.

Empirical evidence from our earlier 100K param experiments:

The TinyStories result is the clearest illustration: the model has 1501 vocabulary tokens but only 3.9K parameters in the transformer layers. It recognises English words but cannot form coherent sentences. Once P_{reason} crosses $\sim 80\text{K}$ the model generates recognisable structural patterns; at 97K it produces fluent domain text.

Implication for the 334K model. At vocab= 256 and $d=88$:

$$P_{\text{reason}} = 334,000 - 256 \times 88 = 334,000 - 22,528 = 311,472$$

The vocabulary tax is only 6.7% of the parameter budget. This is why byte-level Shakespeare is a better fit than word-level datasets at this scale: the transformer layers have 311K parameters to work with, not 4K.

Hardware design implication. A fixed-budget FPGA chip must either specialise to a small-vocabulary domain or separate the embedding onto a dedicated chip (shared-embedding MoE, described briefly in Section 6).

5 Experiments

All experiments were conducted using NeuronFabric release **v1.0.3**. Source code is available at the project repository; checkpoints are reproducible by running the provided training scripts.

Domain	$ V $	Vocab tax	P_{reason}	Final loss
Appointment (GPT-4 synth.)	49	3.1K	96.9K	0.42
MultiWOZ 2.2	302	19.3K	80.7K	2.05
TinyStories	1501	96.1K	3.9K	2.90

Table 5: Vocabulary-budget constraint across three domains at 100K parameter budget ($d=64$). When $P_{\text{reason}} < 20\text{K}$ the model produces recognisable words in incoherent order.

5.1 Correctness Validation

Before any corpus training we validated backpropagation using numerical gradient checks: computing $\partial L/\partial w$ analytically (backward pass) and via finite differences across all components — **NeuronCore**, **AttentionCore**, **AttentionLayer**, and **TransformerBus**. All checks pass. More than a hundred unit and regression tests additionally cover the Adam update rule, BF16W moment accumulation, and the attention softmax backward path independently. The gradient check is the test that cannot be passed by tuning: it computes the same derivative two different ways, and if backpropagation is wrong it fails regardless of how reasonable the output looks.

5.2 Shakespeare Char-Level Training (334K, Adam)

Dataset. The Shakespeare corpus: 1,039,854 training characters, 115,540 validation characters. Byte-level tokenisation, vocab= 256.

Configuration. Architecture as in Table 1. Optimizer: Adam with linear LR schedule, warmup 200 steps, peak LR = 0.003. Training: 80,000 samples (online, batch= 1), GPU accelerated. Checkpoint format: **.neuro** (JSON header + flat binary weights), version-stamped.

Train/validation split. The Shakespeare corpus is split 90/10: 1,039,854 characters for training, 115,540 for validation. The model never sees the validation set during training. All loss and accuracy numbers reported here are measured on the held-out validation set, not the training set — the goal is generalisation, not memorisation.

Variant	Samples	Val loss	Accuracy	ms/sample
GPU Adam FP32	80,000	1.5226	55.79%	19.37
CPU Adam BF16W	80,000	1.5477	54.57%	149.51

Table 6: Shakespeare 334K training results (NeuronFabric v1.0.3, run/results/). GPU Adam FP32 is the oracle baseline. CPU BF16W gap: +0.025 val loss, $7.7\times$ slower — consistent with the precision difference between BF16 and FP32 weight storage.

Training curve. Best eval loss 1.5226 (GPU FP32, at 80K) and 1.5477 (CPU BF16W, at 78K) were reached within the 80K sample budget. Neither variant overfit — eval and train loss tracked closely throughout. Figure 2 shows both curves.

Sample output (CPU Adam BF16W, temperature 0.8, 80K samples):

```
> MENENIUS :
MENENIUS :
if for my life, for a most rever in reason,
And would some a toward had I will bone 't!
```

Second Servant:

Who seem to York at thy lord sede, your longly:
Content, who never he cannot bounds end:
Who's much subject you me shall be short
About and a wife all quarried may may.

The model generates named Shakespearean characters, metered dialogue structure, and contextually coherent responses. This is coherent literary text from a 334K parameter BF16W model trained for ~3.3 hours on CPU (no GPU required).

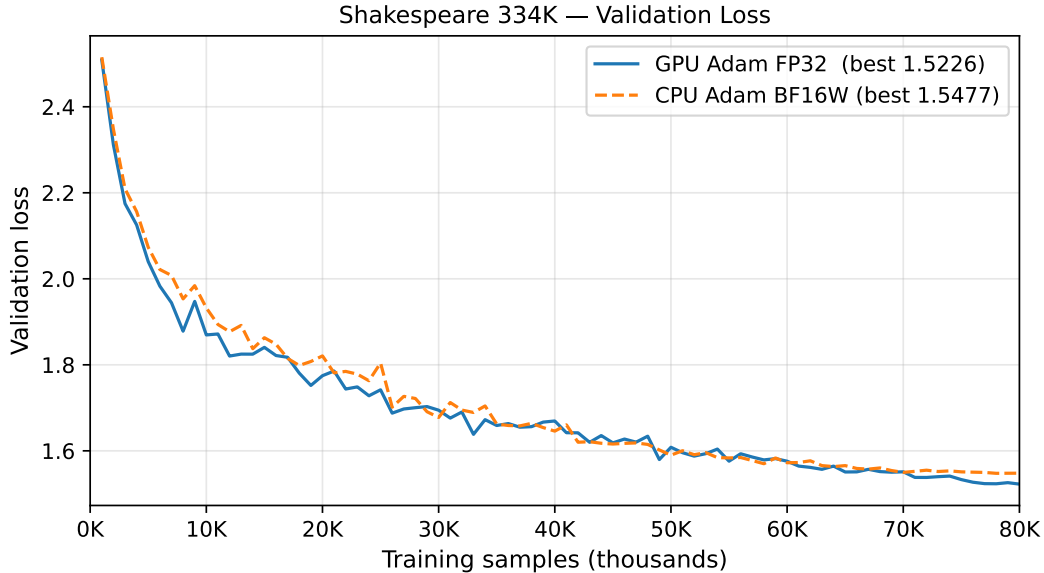


Figure 2: Validation loss vs training samples for Shakespeare 334K model (NeuronFabric v1.0.3). GPU Adam FP32 reaches eval loss 1.5226 at 80K samples (oracle baseline). CPU Adam BF16W reaches 1.5477 — a gap of +0.025, consistent with the precision difference between BF16 and FP32 weight storage.

6 Hardware Feasibility and Next Steps

6.1 FPGA Training Target

The 334K BF16W Shakespeare model is our concrete FPGA training target. No FPGA measurements are included in this paper — we are reporting it here to make the claim falsifiable and to explain why BF16W is necessary rather than optional.

The key properties that make this model FPGA-friendly:

1. **Fits in BRAM.** 3.34 MB weights + moments < 4.0 MB ZCU102 BRAM. No DDR traffic during training.
2. **Batch= 1 online training.** Adam updates one sample at a time. Activation memory scales with one layer at a time (recomputed during backward), not with batch size. Activation recomputation adds compute but no extra SRAM.

3. **No host CPU in the update loop.** The weight update completes on the same SRAM that holds the weights. The host sends a token sequence and receives a loss value. Everything in between happens on chip.

The target result for the FPGA implementation: loss drops from ~ 3.2 (random init) to ~ 1.54 entirely on the FPGA, matching the software result in Table 6, with no host CPU involvement in the Adam update step.

Projected training time on ZCU102 (unverified). At 150–200 MHz (a realistic target for a transformer core with control logic and interconnects) with ~ 600 BF16 FMA units, fully pipelined: ~ 15 – 30 ms/sample. Power: ~ 15 W. These are projections from FPGA characterisation data [10], not measurements. Measured results will be reported in future work.

6.2 Scaling Beyond One Chip

For larger vocabularies or deeper models the vocabulary-budget constraint (Section 4) motivates a multi-chip approach: a dedicated embedding chip pays the vocabulary tax once, and expert chips contain only transformer layers. Inter-chip traffic is the hidden state vector $\mathbf{h} \in \mathbb{R}^{T \times d}$ only — fixed regardless of model depth or total parameter count.

$$\text{bytes per sample} = T \times d \times 4 = 128 \times 88 \times 4 = 45,056 \text{ bytes} \quad (10)$$

This is not a GPU-style gradient bus. Gradients do not cross chip boundaries. Each chip updates its own weights locally. Scaling is achieved by adding chips, not by widening a communication bus.

6.3 Silicon Roadmap

The BF16W scheme is process-independent: 10 bytes per trainable parameter regardless of node. As on-chip SRAM density improves with process scaling, the same architecture accommodates larger models without architectural change. Silicon-level projections are left for future work pending FPGA validation.

The ultimate vision: each chip is an autonomous learning unit. Chips connected via activation links only. No gradient bus. No host CPU. Scaling limited only by the number of chips connected.

7 Discussion

7.1 What This Paper Does and Does Not Claim

To be direct: this paper does not have FPGA measurements. The claim that NeuronFabric can train a transformer entirely on chip rests on software experiments and FPGA architectural analysis, not on a running FPGA system. FPGA validation is future work.

What this paper does establish:

- The backpropagation is mathematically correct (gradient checks, backed by more than a hundred unit and regression tests).
- BF16W Adam converges with a negligible gap ($+0.025$ val loss) versus FP32 Adam on Shakespeare at 80K samples.
- The 334K BF16W model fits in ZCU102 BRAM with 660 KB headroom — this is arithmetic, not a projection.
- The vocabulary-budget constraint is real and measurable across three domains.

7.2 Reproducibility

All results in this paper are reproducible from the published codebase. Every architectural claim is backed by a test that either passes or fails. The gradient check in Section 5 is the hardest to fake — it computes the same derivative two different ways. Training runs are logged to `.log` files alongside each `.neuro` checkpoint, with full hyperparameters and per-sample loss recorded.

To reproduce the Shakespeare 334K results from scratch:

1. `run/1.build.bat` — builds the project (GPU or CPU-only)
2. `run/2.train-gpu-adam-shakespeare.bat` — trains GPU Adam FP32 oracle
3. `run/2.train-cpu-adam-bf16w-shakespeare.bat` — trains CPU BF16W (no CUDA required)
4. `run/3.demo-gpu-adam-shakespeare.bat` — interactive demo from GPU checkpoint
5. `run/3.demo-cpu-adam-bf16w-shakespeare.bat` — interactive demo, CPU BF16W

Checkpoints and logs are saved to `run/results/`. Source code and scripts are available at github.com/Binoculars-X/neuro-fabric.

Acknowledgements

Code repository: github.com/Binoculars-X/neuro-fabric, release tag `v1.0.3`. A preprint of this paper will be deposited on arXiv; a Zenodo/DOI archive will be created at that time to provide a citable, timestamped snapshot of the software and results.

8 Conclusion

We built a complete software reference implementation of a transformer that trains entirely on a single machine, with local Adam, in pure C#, with no external ML framework. We trained a 334K BF16W model on Shakespeare and got coherent text at eval loss 1.5477 — with 660 KB of on-chip SRAM headroom to spare on a mid-range FPGA.

The next step is to run the same training loop on the FPGA and get the same loss curve. If that works, the claim that on-chip transformer training is viable — without a host CPU in the weight update loop — becomes a measured fact rather than a software projection.

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